

6.9 Package thermal characteristics

The maximum chip-junction temperature, $T_J \text{ max}$, in degrees Celsius, can be calculated using the following equation:

$$T_J \text{ max} = T_A \text{ max} + (P_D \text{ max} \times \Theta_{JA})$$

Where:

- $T_A \text{ max}$ is the maximum ambient temperature in °C.
 - Θ_{JA} is the package junction-to-ambient thermal resistance in °C/W.
 - $P_D \text{ max}$ is the sum of $P_{INT} \text{ max}$ and $P_{I/O} \text{ max}$:

$$P_D \text{ max} = P_{INT} \text{ max} + P_{I/O} \text{ max}$$
 - $P_{INT} \text{ max}$ is the product of I_{DD} and V_{DD} , expressed in Watts. This is the maximum chip internal power.
- $P_{I/O} \text{ max}$ represents the maximum power dissipation on output pins:

$$P_{I/O} \text{ max} = \sum (V_{OL} \times I_{OL}) + \sum ((V_{DDIOx} - V_{OH}) \times I_{OH})$$

taking into account the actual V_{OL}/I_{OL} and V_{OH}/I_{OH} of the I/Os at low and high level in the application.

Table 81. Package thermal characteristics

Symbol	Definition	Parameter	Value			Unit
			Junction-ambient Θ_{JA}	Junction-board Θ_{JB}	Junction-case Θ_{JC}	
Θ	Thermal Resistance	LQFP100	39.2	25.1	11.4	°C/W
		LQFP80	42.8	27.1	13.1	
		LQFP64	44.3	26.6	13.5	
		LQFP48	51.4	28.7	16.1	
		LQFP32	51.4	28.7	16.1	
		UFQFPN48	29.9	14.2	12	
		UFQFPN32	40.4	22.3	20.5	

6.9.1 Reference documents

- JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions - Natural Convection (Still Air) available on www.jedec.org.
- For information on thermal management, refer to application note *"Guidelines for thermal management on STM32 applications"* (AN5036) available on www.st.com.